

RESEARCH ARTICLE

Humanity Lifeline: A Resilient Communication and Sensor Network Framework for Disaster Response

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ABSTRACT In the domain of disaster communication, the failure of conventional infrastructures such as cellular towers and power lines poses significant challenges to timely information exchange. Existing solutions often fall short in providing robust and scalable communication networks under such conditions. This study addresses this gap by proposing a hybrid communication system integrating Bluetooth Low Energy (BLE) with LoRa and LoRaWAN technologies. Our approach involves custom-developed hardware: LifeLinks with Renesas RA4W1 microcontrollers and LoRa-Router-Gateways equipped for both peer-to-peer and cloud connectivity. To validate our system, we conducted a series of experiments for high-traffic scenarios, network disruptions, diverse environmental conditions, and coordinated rescue operations. Results demonstrated 99% successful message delivery, minimal data loss, and strong network reliability, indicating the system's efficacy in disaster scenarios. Our research bridges the gap in current literature by offering a comprehensive, resilient communication solution that ensures continuous data flow and robust performance even in severely disrupted environments.

INDEX TERMS Emergency communication, WSN, data aggregation, network robustness, cloud integration.

I. INTRODUCTION

Natural disasters such as earthquakes frequently disrupt critical infrastructure, leaving millions without essential communication services for extended periods. This breakdown severely hampers rescue and relief efforts, as individuals struggle to relay their needs to emergency responders. Traditional networks, such as Global System for Mobile Communications (GSM) and internet services, are often the first to fail, highlighting the necessity for a more resilient and adaptive communication infrastructure [1], [2].

Research in disaster communication has explored various approaches, including satellite communication, mesh networks, and portable base stations. However, these solutions are often constrained by high costs, limited coverage, and reliance on complex infrastructure, which may not be feasible

in all disaster scenarios [3], [4]. Much of the existing work has focused on enhancing long-range communication, with less emphasis on establishing reliable local networks in densely populated urban environments. Additionally, mesh networks using technologies such as Wi-Fi and Zigbee face significant challenges related to range limitations and interference in dense settings [5], [6], [7].

Recent studies have explored innovative methods for improving situational awareness and communication efficiency in disaster zones. For instance, Cai et al. [8] developed a system for real-time environmental identification in borehole rescues by integrating multiple heterogeneous sensors, demonstrating the importance of comprehensive information gathering in specific disaster scenarios. Similarly, [9] Zhang et al. proposed a multi-layer information dissemination model incorporating unmanned aerial vehicles (UAVs), emergency vehicles, and rescuers, while investigating interference optimization strategies to ensure reliable

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communication. These studies highlight the growing recognition of the need for adaptive, multi-layered communication networks capable of handling diverse operational conditions.

Efficiently managing data traffic with varying levels of importance is another critical challenge in disaster response. Sun et al. [10] proposed a game-theoretic approach for multi-priority data transmission in 5G vehicular networks, addressing issues of resource allocation and quality-of-service (QoS) guarantees for applications with different latency constraints. Furthermore, dynamic network function provisioning has been explored as a means of rapidly establishing communication networks in disaster-stricken areas. Sun et al. [11] investigated the use of network function virtualization to enable “Network in a Box” solutions for industrial applications, underscoring the potential of software-defined networks for rapid infrastructure deployment in emergency scenarios.

Despite these advancements, several gaps remain in the literature, particularly regarding scalable, low-power, and cost-effective communication methods that can be quickly deployed in urban environments. There is also a lack of focus on integrating local communication solutions with long-range networks to ensure seamless data transmission to central coordination units.

To comprehensively address these gaps, we propose a hybrid communication system that uses Bluetooth Low Energy (BLE) and LoRa technologies. This system enables individuals to initiate emergency communication by transmitting predefined messages via BLE to LifeLink devices, which then relay the messages to strategically placed LoRa-Router-Gateways. These routers facilitate message transmission to disaster coordination centers, ensuring uninterrupted communication even in the complete absence of traditional network infrastructure. Additionally, the system supports real-time tracking and coordination of rescue teams, improving situational awareness and optimizing emergency resource allocation. We have designed and developed custom hardware, including LifeLinks with Renesas RA4W1 microcontrollers and LoRa-Router-Gateways equipped for both peer-to-peer and cloud connectivity. The LifeLink acts as a crucial aggregator of local data via BLE before relaying it over longer distances using LoRa. Thus our study proposes first hybrid system using both LoRa and LoRaWAN for resilient disaster communication. Unlike many existing studies that rely primarily on simulations, we have conducted a series of comprehensive real-world experiments performing high-traffic scenarios, network disruptions, diverse environmental conditions, and coordinated rescue operations within a 5 km x 5 km test area. These experiments provide empirical validation of the system’s efficacy under realistic disaster conditions.

Initial experiments demonstrate the system’s effectiveness in optimizing transmission latency and network traffic, highlighting its potential as a scalable, cost-efficient, and resilient solution for disaster communication. By integrating local and long-range communication strategies, our approach

aims to bridge critical gaps in disaster response, ultimately enhancing the efficiency and effectiveness of emergency operations. Furthermore, large-scale real-world tests conducted in Istanbul, a city with a population of 25 million, validate the system’s scalability and robustness in dense urban environments.

The paper is organized as follows: Section II reviews related works. Section III outlines the methods employed. In Section IV, we describe our experimental platform, the experiments conducted and the results. Finally, Section V provides the conclusion and discussion.

II. RELATED WORK

The development of communication technologies for emergency scenarios is a rapidly evolving field. Recent advancements in wireless communication technologies, particularly Low-Power Wide-Area Networks (LPWAN), have shown significant promise in enhancing communication capabilities during critical situations [12], [13].

Vithayathil et al. [14] proposed a LoRa-based network for disaster rescue operations, combining LoRa and Wi-Fi technologies to enable emergency messaging during catastrophic events. Similarly, Macaraeg et al. [15] introduced a LoRa-based mesh network using a modified AODV routing protocol for reliable communication in disaster-prone areas. Sangrit et al. [16] explored LoRa communication in mountainous regions, while Höchst et al. [17] developed a low-cost, long-range device-to-device communication method using LoRa for crisis situations.

Sciullo et al. [5] introduced LOCATE, an emergency communication system using LoRa for multi-hop dissemination of requests in environments without cellular connectivity. Yen et al. [18] discussed using LoRa for temporary network infrastructure in disaster relief, and Manuel et al. [19] propose a LoRa-based architecture for search and rescue missions. Baumgärtner et al. [20] presented LoRAgent, a DTN system using LoRa for disaster communication, and Dubey et al. [21] designed a LoRa-based model for disaster scenarios with compromised infrastructure.

Further advancements included Imran et al. [22]’s LoRa-based flood rescue system integrating IoT and GPS, and Cardenas et al. [23]’s low-cost messaging system for underserved areas using LoRa. Li et al. [24] introduced AWCT, a communication framework enhancing IoT resilience during disasters, and Ingale et al. [25] proposed a Distressed Positioning System using LoRa. Lastly, Karthikeyan et al. [26] presented a LoRa-based device for disaster and fleet management, and Li and Chen [27] introduced BLE2LoRa for cross-technology communication between BLE and LoRaWAN.

Jiang et al. [28] proposed a hybrid LPWAN mesh network architecture that combines LoRa for long-range and ANT for short-range communication to enhance coverage and reduce power consumption for IoT devices. It introduced a centralized TDMA scheduling protocol for the LoRa mesh

network to ensure collision-free communication and presents performance results from real-world deployments.

Ayele et al. [29] presented a new dual-radio IoT network architecture for wildlife monitoring systems (WMS). This system dynamically switches between BLE for short-range communication within animal herds and LoRa for long-range communication to a gateway, optimizing energy consumption based on animal proximity. The paper also provides an analytical model demonstrating the improved energy efficiency of this hybrid approach compared to using LPWAN alone.

Shilpa et al. [30] presented the design and implementation of a hybrid LPWAN architecture using LoRa for long-range and Wi-SUN for a low-latency mesh network to achieve better coverage, capacity, and lower power consumption for IoT applications. The main contribution was the successful integration of these two LPWAN technologies, validated through a real-world deployment of a smart street light control system.

Existing studies demonstrate progress in developing communication technologies for disaster scenarios, as well as advancements in sensor networks and UAV applications. However, a critical gap remains in the literature regarding the comprehensive evaluation of hybrid systems that integrate multiple communication technologies to enhance robustness and reliability in dynamic environments. Specifically, existing studies do not dynamically adjust the spreading factor to adapt to changing network topologies, potentially underestimating its energy optimization capabilities in real-world dynamic scenarios. This study addresses this gap by proposing a novel hybrid communication framework that incorporates adaptive mechanisms to optimize communication parameters in response to evolving network conditions. Through simulations and real-world experiments, this research aims to offer a more resilient solution for critical communication during emergencies.

Despite the advancements presented in existing studies, several limitations warrant consideration. For instance, [28] introduced a hybrid LoRa and ANT network featuring an Adaptive LoRa Link (ALL), yet its scope remained restricted to transmission power adjustments, overlooking the potential benefits of dynamically altering the spreading factor. Additionally, the reliance on flooding the complete TDMA schedule table across the entire network poses scalability challenges and inefficiencies during the setup phase due to LoRa's packet size constraints. Similarly, [29] acknowledged the practical complexities of their dual-radio BLE and LoRa wildlife monitoring system, noting that the modeled wireless parameters may not be directly translatable to physical hardware. Their analysis primarily relied on path-loss-based range estimations, and they emphasized the necessity for future research involving real-world validation of their simulation results, alongside the exploration of online proximity estimation techniques and practical performance testing of LoRa in the target environment.

Moreover, [30], while demonstrating a functional hybrid LoRa and Wi-SUN architecture for smart street lighting,

initially implemented LoRa in a point-to-point configuration rather than the intended star topology, potentially limiting the generalizability of their findings. Their future work also highlights the need for scaling up the deployment and further optimizing the architecture by incorporating additional considerations such as power consumption and latency.

Multi-sensor data fusion has emerged as a powerful technique for gaining comprehensive insights in disaster environments. Cai et al. [8] presented a borehole rescue system that utilizes the fusion of data from gas, audio, video, and personnel positioning sensors, achieving high accuracy in identifying risks in underground disaster areas. While this work highlights the benefits of data fusion in a specific context, our research focuses on a more broadly applicable communication infrastructure utilizing LPWAN technologies for diverse disaster scenarios.

Game-theoretic approaches have been explored to optimize resource allocation and manage diverse traffic requirements in various networks. Sun et al. [10] presented a game-theoretic framework for joint mode selection and power adaptation in 5G vehicular networks to prioritize data transmission based on latency constraints. While our work shares the goal of efficient resource utilization, it focuses on using the characteristics of LPWAN technologies in a dedicated post-disaster communication system.

The 'Network in a Box' concept aims to provide rapidly deployable communication infrastructure. Sun et al. explored the use of Network Function Virtualization (NFV) for dynamic network function provisioning in industrial applications, enabling flexible and software-based network deployments. In contrast, our work focuses on a dedicated hardware and LPWAN-based solution to the unique challenges and requirements of post-disaster communication.

The challenges of information dissemination and interference management in disaster communication networks with diverse nodes have been studied by several researchers. Zhang et al. [9] developed a multi-layer information dissemination model for communication networks in disaster areas, considering UAVs, emergency vehicles, and rescuers. Their work focused on analyzing the information propagation process and proposing an interference optimization algorithm based on convex optimization to minimize network deployment costs. Our proposed system also addresses communication in disaster scenarios but with a different architectural approach, using LPWAN technologies and focusing on enabling direct communication between affected individuals and emergency responders through a hybrid network infrastructure.

Despite the significant advancements in post-disaster communication, existing studies exhibit several common limitations. First, while numerous works propose hybrid communication architectures integrating multiple technologies (e.g., LoRa, BLE, Wi-Fi, and Wi-SUN), they often lack adaptive mechanisms that dynamically adjust key communication parameters, such as the spreading factor, transmission

power, and routing strategies, in response to evolving network conditions. This omission limits the energy efficiency and scalability of the proposed solutions, particularly in dynamic and resource-constrained disaster environments. Second, many studies rely on simulation-based evaluations or theoretical models that do not fully account for real-world constraints such as hardware limitations, environmental interference, and unpredictable user mobility patterns. As a result, their findings may not be directly applicable to large-scale deployments. Third, existing hybrid communication frameworks often focus on specific application domains (e.g., wildlife monitoring, vehicular networks, or smart infrastructure) without addressing the unique challenges posed by disaster scenarios, where infrastructure is severely damaged, network topology is highly dynamic, and energy efficiency is critical. Moreover, certain approaches rely on centralized scheduling mechanisms, such as TDMA-based coordination, which can introduce scalability bottlenecks and excessive setup overhead in large-scale disaster areas.

In contrast, our study introduces a novel hybrid communication framework that overcomes these limitations by seamlessly integrating BLE, LoRa, and LoRaWAN under a unified system with high efficiency and reliability. In contrast to previous works with predominantly static configurations, our approach incorporates an adaptive mechanism that dynamically optimizes communication parameters based on real-time network conditions, thereby enhancing both energy efficiency and communication reliability. Our study presents the first hybrid approach that uses both BLE, LoRa and LoRaWAN to establish a robust and resilient disaster communication network. Additionally, our work is one of the first to conduct large-scale real-world field tests in a dense urban environment, specifically within the 25-million-population metropolis of Istanbul. These experiments provide empirical validation of our system's robustness and scalability under realistic disaster conditions. Furthermore, unlike centralized scheduling approaches, our system employs a decentralized and lightweight coordination strategy, reducing overhead and improving scalability. By addressing these critical gaps, our research contributes to the development of a more resilient and adaptive communication infrastructure for emergency response and disaster recovery operations.

III. PROPOSED APPROACH: HUMANITY LIFELINE

This section details the proposed a robust hybrid communication system (Humanity Lifeline) designed for resilient post-disaster communication. The system's network architecture comprises several key components working in concert to ensure reliable communication when conventional networks are disrupted: smartphones with BLE used by affected individuals to initiate communication; LifeLink devices strategically positioned to aggregate and forward data received via BLE; LoRa-Router-Gateways that relay data over long distances, either to the cloud via LoRaWAN or through peer-to-peer LoRa connections; a centralized Cloud Server hosted on AWS for data processing and coordination;

and Emergency Response Units (ambulances and search and rescue teams) that receive processed data for action.

The design of the Humanity LifeLine system envisions a flexible and scalable network. The number of BLE-enabled smartphones (user devices) and LifeLink devices would adapt to the scale of the disaster and the distribution of affected individuals, with the system designed to support a potentially large number of BLE connections to individual LifeLinks.

To evaluate the system's performance under realistic conditions, our experiments involved a specific deployment within a 5 km x 5 km test area in Istanbul, Turkey (a city of 25 million population). This setup included 100 LifeLink devices strategically positioned to represent the potential distribution of affected individuals, collecting and aggregating data transmitted from 500 mobile phones simulating users sending emergency messages. Furthermore, 5 LoRa-Router-Gateways were deployed across the test area to ensure adequate coverage and evaluate the system's long-range communication capabilities. These gateways were configured to operate in both LoRaWAN and peer-to-peer LoRa modes.

Performing the complementary strengths of BLE and LoRa technologies, Humanity LifeLine establishes a reliable communication infrastructure when conventional networks are disrupted. The system facilitates communication between user smartphones and dedicated LifeLink devices via BLE, which then relay data over long distances to LoRa-Router-Gateways using LoRa. Depending on network availability, data is transmitted to a cloud server via LoRaWAN or relayed through a peer-to-peer LoRa mesh network to reach disaster coordination centers, ambulances, and search and rescue teams. The subsequent subsections will detail the system architecture and design, hardware development, and the experimental evaluation of this proposed approach.

A. SYSTEM ARCHITECTURE AND DESIGN

1) HYBRID COMMUNICATION FRAMEWORK

Humanity LifeLine framework integrates multiple communication layers to ensure reliability and efficiency:

- **BLE Layer:** Enables short-range, low-power communication between smartphones and LifeLink, ensuring real-time messaging in disaster scenarios.
- **LoRa Layer:** Extends communication range by transmitting data from LifeLink to LoRa-Router-Gateway, maintaining connectivity in disrupted areas.
- **LoRaWAN Layer:** Facilitates cloud connectivity, enabling real-time data aggregation and dissemination for emergency coordination.

2) COMMUNICATION FLOW

The system follows a structured data flow for efficient disaster response:

Local Communication (Bottom-Up): Users send messages via BLE to LifeLink, which aggregates and forwards



FIGURE 1. Data exchange between devices.

data via LoRa to LoRa-Router-Gateway. This ensures immediate communication even when conventional networks fail.

Intermediate Communication: If connected to LoRaWAN, the LoRa-Router-Gateway transmits data to the cloud. If not, it relays messages to other LoRa-Router-Gateways via peer-to-peer LoRa, ensuring network continuity.

Data Processing and Distribution (Top-Down): The cloud processes received data, making it accessible to disaster coordination centers. Emergency teams, such as ambulances and search-and-rescue units, receive relevant updates for effective response. Figure 1 illustrates the simplified data exchange process.

Data Packet Structure: The proposed communication system has two distinct packet structures for LoRa and LoRaWAN transmissions. The LoRaWAN packet structure is designed for efficient long-range communication, whereas the LoRa packet structure is optimized for inter-gateway data exchange.

LoRaWAN Packet Structure: The LoRaWAN packet structure is defined as follows:

```
typedef struct {
    uint32_t id : 28;
    uint32_t pkt_cnt : 4;
    uint32_t date_time : 20;
    uint32_t data_type : 12;
    uint32_t data : 24;
} RECORD_LORAWAN_t;
```

LISTING 1. LoRaWAN packet structure.

Each field in the LoRaWAN packet is defined as follows:

- **id** (28 bits): Unique identifier for the node.
- **pkt_cnt** (4 bits): Packet counter for tracking message sequences.
- **date_time** (20 bits): Timestamp of data transmission.
- **data_type** (12 bits): Specifies the type of data being transmitted.
- **data** (24 bits): Actual payload data.

LoRa Packet Structure: For inter-gateway communication, the LoRa packet structure is divided into a header and a payload, as defined below:

```
typedef struct {
    uint32_t id : 28;
    uint32_t pkt_cnt : 4;
    uint32_t date_time : 20;
} RECORD_HEADER_t;

typedef struct {
    uint32_t data_type : 12;
    uint32_t data : 24;
} RECORD_PAYLOAD_t;

typedef struct {
    RECORD_HEADER_t header;
    RECORD_PAYLOAD_t payload[MAX_PAYLOADS];
    uint8_t payload_count;
} RECORDS_t;
```

LISTING 2. LoRa packet structure.

The fields in the LoRa packet structure are defined as follows:

- **id** (28 bits): Unique identifier for the transmitting node.
- **pkt_cnt** (4 bits): Packet counter.
- **date_time** (20 bits): Timestamp of the message.
- **data_type** (12 bits): Type of transmitted data.
- **data** (24 bits): Actual data being sent.

Example Data Packets: The following example illustrates data transmission from a node using LoRaWAN, followed by a compressed inter-gateway LoRa message:

LoRaWAN to LoRa Message via MQTT:

```
Received -> 00 49 3E A8 01 5F C0 69 00 00 01 00
49 3E A9 01 5F C0 68 00 00 01 00 49 3E AA 01
5F C0 05 06 1F 52 00 49 3E AB 01 5F C0 07 04
0D CE
```

LoRa Inter-Gateway Message (Compressed Packet)

```
Queue -> 00 49 3E A8 01 5F C0 01 00 0C 0B D2 00
69 00 00 01 00 68 00 00 01 00 05 06 1F 52 00
07 04 0D CE
```

The message structure for the LoRa inter-gateway transmission is detailed as follows:

- **Header:** 00 49 3E A8 01 5F C0 01 00
- **Gateway ID:** 0C 0B D2
- **Message Code + Personal ID:** 00 69 00 00 01
- **Message Code + Message Type:** 00 68 00 00 01
- **Message Code + Latitude:** 00 05 06 1F 52
- **Message Code + Longitude:** 00 07 04 0D CE

This structured approach enables efficient disaster communication by reducing message size while maintaining essential information, ensuring robust data transmission in emergency scenarios.

3) HARDWARE COMPONENTS

- **Smartphones with BLE:** User devices for initiating communication.
- This hybrid approach ensures robust disaster communication, bridging short-range BLE connectivity with long-range LoRa transmission. The system's scalable and resilient architecture enhances disaster

coordination, facilitating timely and reliable communication even in infrastructure-compromised scenarios. By integrating both local and long-range transmission, the proposed solution strengthens emergency response and improves overall disaster management efficiency.

B. HARDWARE DEVELOPMENT

In this study, we have developed a robust hardware architecture designed to facilitate reliable communication during emergency scenarios. The primary components of this architecture include mobile phones, LifeLinks, LoRa-Router-Gateways, and cloud infrastructure, each playing a critical role in ensuring seamless data transmission and coordination. The major hardware devices LifeLink and LoRa-Router-Gateway are designed and developed 100% in house during this study.

1) MOBILE PHONES

In our emergency communication system, mobile phones act as the initial interface for users in distress. These devices are equipped with BLE technology, a standard for wireless communication that operates in the 2.4 GHz ISM band. BLE is chosen for its low power consumption and ability to maintain connections over a short range, typically up to 100 meters in optimal conditions. When an emergency occurs, users can send predefined messages from their mobile phones to the LifeLink via BLE. These messages contain critical information such as the user's location, status, and specific help requests. The choice of BLE is strategic due to its low energy requirements, ensuring that even in power-constrained situations, mobile phones can reliably communicate with the LifeLink.

2) LIFELINK

The LifeLink is a crucial element in our disaster communication system, designed in-house to aggregate data from the disaster zone. At its core is the Renesas RA4W1 microcontroller, featuring a 32-bit ARM Cortex-M4 running at 48 MHz, with integrated Bluetooth 5.0, 512 KB Flash memory, and 96 KB SRAM, ideal for processing data from multiple devices. Based on the user requirements, different LifeLink devices are designed and developed. These devices are introduced in Figure 2. Board layout and block diagram are presented in Figure 3.

Bluetooth 5.0 on the RA4W1 enhances data throughput, range, and interference resilience, ensuring stable communication in emergencies. The microcontroller's power efficiency supports extended operation on battery power.

The LifeLink also includes a LoRa module with a Semtech SX1276 transceiver, operating in the sub-GHz bands (868 MHz in Europe, 915 MHz in North America), enabling long-range communication of up to 15 km in rural areas and 5 km in urban settings. This dual technology setup allows the gateway to efficiently process and forward data, optimizing both short- and long-range communication in the emergency network.



FIGURE 2. LifeLink device views.

TABLE 1. LifeLink device technical specifications.

Feature	Specification
Microcontroller	Renesas RA4W1 (ARM Cortex-M4, 48 MHz)
Bluetooth	Bluetooth 5.0
Memory	512 KB Flash, 96 KB SRAM
LoRa Transceiver	Semtech SX1276
LoRa Frequency Band	868 MHz (EU) / 915 MHz (NA) (veya kullanılan diğer frekanslar)
LoRa Range	Up to 15 km (rural), 5 km (urban)
Power Source	3.7V Li-ion Battery (1000 mAh)
Operating Temperature	-20°C to +60°C
Dimensions	80 mm x 50 mm x 20 mm
Weight	50 g
Antenna	Integrated PCB Antenna (BLE: 0 dBi, LoRa: 2 dBi)
Sensors	GPS, Accelerometer, Temperature Sensor, Vibration, Sound

The combination of BLE and LoRa in the LifeLink ensures that data from user devices is aggregated and forwarded efficiently. The gateway processes incoming BLE messages, converts them into a format suitable for LoRa transmission, and forwards them to the LoRa-Router-Gateway. This multi-technology approach optimizes both short-range and long-range communication, making the LifeLink a versatile and robust component in the emergency communication network.

The LifeLink device is a crucial component of the Humanity LifeLine system, responsible for aggregating data from user smartphones and relaying it to the LoRa-Router-Gateways. To provide a clear overview of its capabilities, the key technical specifications of the LifeLink device are summarized in Table 1.

3) LORA-ROUTER-GATEWAY

LoRa-Router-Gateways are vital for connecting LifeLinks to the cloud, equipped to handle both

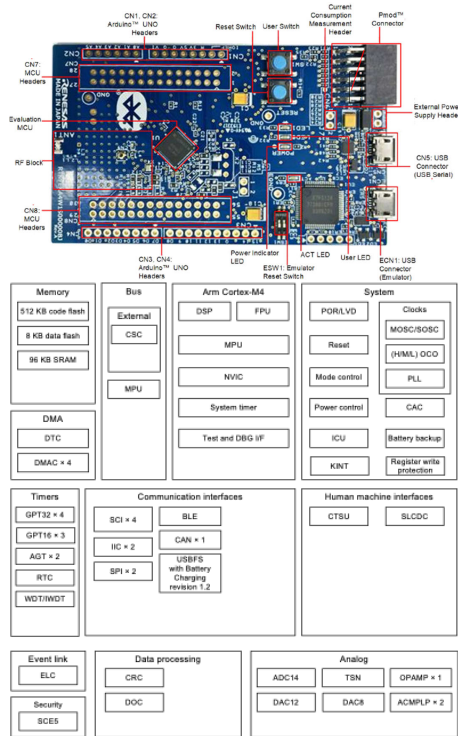


FIGURE 3. Board layout and block diagram - hardware specifications. Source: <https://www.renesas.com/en/document/man/renesas-ra4w1-group-users-manual-hardware?r=1054151>.

intra-network communication and cloud interfacing. Each router (See Figure 4) features a Semtech SX1301 digital baseband chip, capable of managing complex packet routing and supporting multiple spreading factors and channels simultaneously, allowing it to handle numerous devices and data rates.

High-gain LoRa antennas enhance signal strength and range, optimized for the specific frequency bands used by LoRa technology. Operating in the sub-GHz frequency range, these routers achieve long-range communication with minimal interference.

Integrated with LoRaWAN, these routers support secure, low-power, wide-area networking, enabling efficient and secure data transmission from the disaster zone to the cloud. Additionally, they feature a fallback mechanism that allows peer-to-peer communication between routers when direct LoRaWAN network access is unavailable, ensuring network resilience in challenging conditions. Finalized gateway device views are introduced in Figure 5 and deployed devices on the field are presented in Figure 6.

C. CLOUD INFRASTRUCTURE, CONTROL CENTER, AND SECURITY

1) CLOUD INFRASTRUCTURE

The core communication system involves the transmission of emergency messages from user smartphones via BLE to LifeLinks, which then relay this data over long distances using LoRa to the LoRa-Router-Gateways.

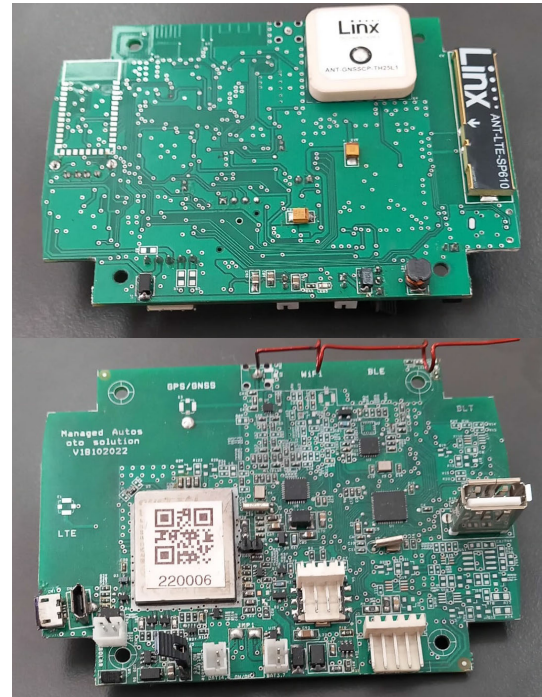


FIGURE 4. LoRa-Router-Gateway hardware views - sides 1-2.



FIGURE 5. LoRa-Router-Gateway hardware views with LORA antenna.

These gateways serve as the bridge between the field devices and the cloud infrastructure.

When a LoRa-Router-Gateway has LoRaWAN connectivity, it transmits the aggregated data to the AWS cloud via AWS IoT Core. AWS IoT Core acts as the entry point for device data into the AWS ecosystem, managing secure connections and communication.



FIGURE 6. Deployed LoRa-Router-Gateway devices.

Upon arrival in AWS IoT Core, the data is then routed to various other AWS services for processing, storage, and analysis. The interaction of the specifically mentioned services (S3, DynamoDB, and Kinesis) with this incoming data stream is as follows:

Amazon Kinesis is crucial for capturing and processing the real-time stream of data coming from the LoRa-Router-Gateways. This allows for immediate analysis of incoming emergency messages, enabling near real-time situational awareness of the disaster zone.”

AWS Lambda, our serverless computing service, is triggered by the arrival of new data in Kinesis or IoT Core. Lambda functions perform essential tasks such as data transformation, validation, and initial analysis of the emergency messages. This processed data is then often directed to other services.

Amazon DynamoDB is used to store metadata and real-time data that requires quick access and retrieval. This can include the current status and location of affected individuals or the operational status of the network devices. The low latency of DynamoDB allows the control center UI to quickly access and display critical, up-to-the-minute information to disaster management teams.

Amazon S3 (Simple Storage Service) provides durable and scalable storage for the bulk of the data collected from the disaster zone. This includes historical records of emergency messages, sensor data logs, and other information that needs to be retained for analysis and reporting.

The processed and stored data in these AWS services is then utilized by the control center UI to provide real-time dashboards, alert notifications, and historical data analysis, enabling disaster management teams to make informed decisions and coordinate response efforts effectively.

Furthermore, Amazon RDS manages structured data for generating reports and analytics, drawing upon the data stored in S3 and potentially aggregated and processed by Lambda.

Amazon SNS and SQS facilitate messaging and communication between the various microservices within the AWS architecture, ensuring reliable data flow and notifications to relevant entities, including emergency responders.

In essence, the AWS services act as the intelligent backend of our system, taking the raw data transmitted through the BLE and LoRa layers and transforming it into actionable information that can be used to enhance disaster response. The real-time processing capabilities of Kinesis, the rapid data access of DynamoDB, and the durable storage of S3 are all critical components in ensuring the system’s efficacy in emergency scenarios.

2) CONTROL CENTER UI

The control center user interface (UI) is designed to provide disaster management teams with a responsive, user-friendly tool for real-time analysis, reporting, and action. It integrates with AWS cloud services, accessing data from Amazon S3 and AWS IoT Core for live monitoring and visualization. Key features include real-time dashboards, alert notifications, and historical data analysis, enabling teams to assess response efforts and make informed decisions. The UI also offers reporting capabilities for generating detailed disaster response reports.

3) SECURITY

Addressing the crucial aspects of data security and the specifics of our testing procedures, proposed framework incorporates robust mechanisms to ensure data integrity and privacy without compromising the efficiency of LoRa communication [31], [32]. Data security in the LoRa communication layer is achieved through the implementation of AES CTR (Advanced Encryption Standard Counter Mode) encryption [33], [34]. This method allows for the encryption of data packets without increasing their size, which is a critical consideration in low-bandwidth networks like LoRa. By using AES CTR, we ensure the confidentiality and integrity of the sensitive information being transmitted across the network, thereby addressing potential concerns regarding data privacy.

In conclusion, our hardware architecture, comprising mobile phones, LifeLinks, LoRa-Router-Gateways, and cloud infrastructure, is designed to provide a reliable and efficient communication network for emergency scenarios. Each component is optimized for its specific role, ensuring seamless data transmission, robust performance, and effective disaster management.

IV. EXPERIMENTS

To evaluate the performance and reliability of our emergency communication system, we conducted real-world comprehensive tests involving various experimental setups and scenarios. These tests aimed to ensure the system’s efficacy during actual disaster situations.

A. EXPERIMENT SETUPS

Our experimental setup combined hardware and software for disaster conditions. Smartphones with BLE sent predefined emergency messages to LifeLink devices built with the Renesas RA4W1 microcontroller. These messages were then

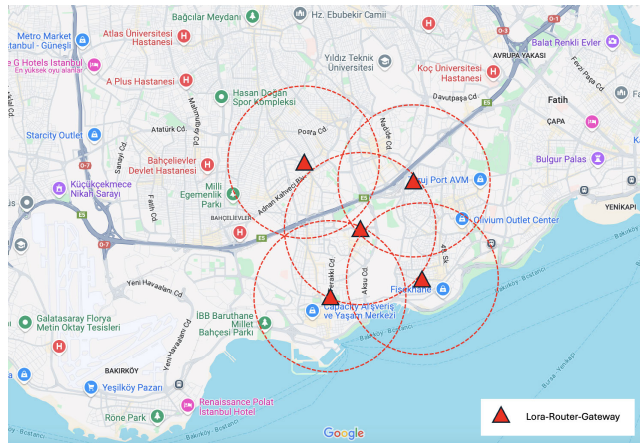


FIGURE 7. Deployed lora-router-gateway.

forwarded to LoRa-Router-Gateways, which relayed data to AWS cloud infrastructure via LoRaWAN or peer-to-peer LoRa communication.

The experiments were conducted within a $5 \text{ km} \times 5 \text{ km}$ test area, representing a realistic disaster zone. We deployed 100 LifeLink devices to collect and aggregate data from 500 mobile phones, simulating affected individuals. This included 25 physical smartphones and 475 emulated devices running on computers near the LifeLink units. Five LoRa-Router-Gateways were positioned strategically to ensure coverage, operating in both LoRaWAN mode (for cloud connectivity via AWS IoT Core) and peer-to-peer mode (for network disruptions). The gateway distribution allowed for overlapping coverage and multi-hop communication, ensuring data transmission even in cases of direct cloud connectivity loss. Figure 7 visually illustrates the deployment of the LoRa-Router-Gateways within the $5 \text{ km} \times 5 \text{ km}$ test area which is home to a population of 219,893.

This setup enabled evaluation of key performance metrics such as range, coverage, data throughput, and resilience under varying network loads. Additionally, tests were conducted in both line-of-sight and non-line-of-sight conditions to assess system performance in diverse environments, including urban and rural settings. Apart from the emulated mobile devices, the entire setup was tested in real-world conditions.

B. EXPERIMENT SCENARIOS AND EVALUATION METRICS

Several experiment scenarios were designed to evaluate the system's performance under different conditions. These scenarios were intended for potential real-world situations and challenges that the system might encounter during a disaster. Summary of chosen experimental scenarios is given in Table 2.

In the High Traffic Handling scenario, message throughput was measured as the number of emergency messages successfully processed within 10 minutes, while data loss was calculated by comparing sent and received messages. Latency was assessed through timestamping at the source (smartphone) and destination (cloud server or emergency

teams), computing end-to-end and average latency over multiple transmissions.

For Network Robustness, the peer-to-peer transmission success rate was determined by the percentage of messages relayed between LoRa-Router-Gateways during LoRaWAN disruptions. Recovery time referred to the duration required to restore communication. Data accuracy ensured message integrity, while information accuracy validated correct data processing for emergency teams.

Environmental impact was analyzed through success rates in indoor, outdoor, and obstructed settings. Response time measured the delay between cloud processing and data reception by emergency teams. Scalability testing examined latency and data loss under varying user loads. Battery life was monitored across different usage conditions until depletion.

Interference testing compared latency and data loss in high-electromagnetic and low-interference environments. Range testing determined the maximum stable communication distance for BLE and LoRa. Fault tolerance was assessed by inducing hardware failures and observing the effects on message delivery. Multi-hop communication performance was evaluated by measuring latency and data loss across different numbers of LoRa relays.

Performed experiments and the results are introduced respectively in Table 3 and in Table 4. The results from the various testing scenarios present the disaster communication system's effectiveness, reliability, and robustness. In the high traffic handling experiment, the system demonstrated an impressive ability to manage a substantial volume of emergency messages, with a 99% success rate in message reception by the LifeLink and an overall data delivery success rate of 98.8%. The average message latency of 1.5 seconds is well within acceptable limits, confirming the system's capability to handle multiple simultaneous messages efficiently.

Network robustness and reliability were validated disruptions in LoRaWAN connectivity. The system showed a high success rate of 94% in peer-to-peer transmissions with an average recovery time of 5 seconds, and a minimal data loss of 0.2% during disruptions. This robustness is crucial for maintaining communication during network instability, ensuring reliable data transmission in emergency situations.

Response time and data integrity tests further affirmed the system's efficiency, with an end-to-end latency of 1.5 seconds and 100% data accuracy, indicating that the system maintains both speed and reliability in data transmission.

The evaluation of performance under various environmental conditions showed high success rates, particularly in outdoor environments (99.5%) and reasonably good performance in obstructed environments (95%), despite some signal attenuation. This versatility ensures the system's reliability across different physical settings.

Finally, the comprehensive test of coordinated rescue operations demonstrated the system's capability to process and analyze data quickly and accurately, with a 99.5%

TABLE 2. Summary of experiment scenarios.

Experiment Scenario	Objective	Description
High Traffic Handling	Evaluate system's capacity for multiple messages	heavy user traffic to assess the ability of LifeLinks and LoRa-Router-Gateways to manage data volume.
Network Robustness	Test fallback during connectivity issues	LoRaWAN disruptions and evaluated peer-to-peer LoRa-Router-Gateway communication for continuity.
Response Time	Measure latency and data accuracy	Tracked latency from BLE message to cloud processing, ensuring quick and accurate data transmission.
Environmental Conditions	Assess performance in varied settings	Tested system in different environments to evaluate resilience and signal penetration.
Rescue Operations	Validate system end-to-end	Full operation from data collection to responder coordination, ensuring overall functionality.
Scalability Testing	Evaluate performance with more users	Varying user loads to assess system scalability and identify bottlenecks.
Battery Life	Measure power consumption	Tested battery usage of devices under different conditions to ensure long-term operation.
Interference Testing	Assess performance in interference-heavy areas	Tested reliability in environments with high electromagnetic interference, like urban zones.
Range Testing	Determine communication range	Tested BLE and LoRa ranges in different settings to optimize deployment.
Heavy Load Latency	Analyze latency under heavy traffic	Evaluated latency during high traffic during major disaster scenarios.
Fault Tolerance	Test response to hardware failures	Assessed system handling of hardware failures and redundancy effectiveness.
Multi-Hop Communication	Test multi-hop LoRa performance	Evaluated reliability of multi-hop LoRa communication for data routing.

TABLE 3. Summary of experiment results - Part 1.

Scenario	Metric	Result
High Traffic Handling	Message Throughput	5,000 messages/10 minutes
	Data Loss	1.2%
	Average Latency	1.5 seconds
Network Robustness	Peer-to-Peer Transmission Success	94%
	Average Recovery Time	5 seconds
	Data Loss During Disruptions	0.2%
Response Time and Data Integrity	End-to-End Latency	1.5 seconds
	Data Accuracy	100%
Environmental Conditions	Indoor Success Rate	98%
	Outdoor Success Rate	99.5%
	Obstructed Success Rate	95%
Coordinated Rescue Operations	Processed Messages	1,000
	Response Time to Teams	2 seconds
	Information Accuracy	99.8%

success rate in message processing and a 2-second response time to emergency teams. The accuracy of the disseminated information was exceptionally high at 99.8%, underscoring

TABLE 4. Summary of experiment results - Part 2.

Experiment	Condition	Result	Observation
Scalability	100 Users	120 ms latency, 1.2% loss	Stable
	500 Users	160 ms latency, 2.8% loss	Acceptable
	1000 Users	200 ms latency, 5.5% loss	Degraded
	2000 Users	280 ms latency, 9.3% loss	Critical
Battery Life	Idle	72h (gateway), 96h (router)	Optimal
	Moderate Usage	48h (gateway), 72h (router)	Acceptable
	Heavy Usage	24h (gateway), 36h (router)	Degraded
Interference	Urban	180 ms latency, 3.5% loss	Acceptable
	Industrial	220 ms latency, 4.8% loss	Degraded
Range	Urban	BLE 30m, LoRa 5km	Limited
	Suburban	BLE 50m, LoRa 10km	Good
	Rural	BLE 70m, LoRa 15km	Optimal
Latency	500 Users	180 ms latency	Stable
	1000 Users	240 ms latency	Degraded
	2000 Users	320 ms latency	Critical
Fault Tolerance	Single Failure	50 ms delay, 0% loss	Reliable
	Multiple Failures	150 ms delay, 1.5% loss	Resilient
Multi-Hop	2 Hops	180 ms latency, 1.2% loss	Stable
	3 Hops	240 ms latency, 2.4% loss	Acceptable
	4 Hops	300 ms latency, 4.1% loss	Degraded

the system's potential to facilitate timely and effective communication with emergency responders.

Overall, the experiments confirm that the system is highly effective for disaster communication, offering reliable, robust, and efficient performance across various scenarios. The system's scalability, quick recovery from disruptions, low latency, and high data integrity make it a suitable solution for enhancing emergency response operations.

V. CONCLUSION AND DISCUSSION

In the domain of post-disaster communication systems, the disruption of conventional telecommunication and infrastructure poses significant challenges to timely and efficient rescue operations. Existing literature highlights the need for robust, scalable, and reliable communication networks that can function independently of damaged infrastructure. Despite various approaches using LoRa, BLE, and hybrid networks, gaps remain in achieving seamless integration, optimal performance under high traffic, and reliability in diverse environmental conditions.

Our proposed hybrid communication system addresses these gaps by combining the strengths of BLE and LoRa technologies. Proposed system presents the first hybrid approach that uses both BLE, LoRa and LoRaWAN to establish a robust and resilient disaster communication network. The system

features smartphones as user devices, LifeLinks equipped with Renesas RA4W1 microcontrollers for BLE communication and LoRa antennas, and LoRa-Router-Gateways that facilitate peer-to-peer communication and connectivity to the cloud via LoRaWAN. The cloud infrastructure, hosted on AWS, enables centralized data processing and dissemination, ensuring that critical information reaches disaster coordination centers and emergency responders efficiently.

To validate our approach, we performed extensive experiments by real-world deployments. Real-world tests presented the system's performance. The experiment results indicate that our hybrid communication system is well-suited for disaster scenarios, offering a reliable, scalable, and efficient solution for maintaining communication when traditional infrastructure fails.

Humanity Lifeline system proposes hybrid communication system achieves exceptional performance in terms of reliability, scalability, and robustness under various disaster scenarios. Its unique integration of LoRa and LoRaWAN technologies differentiates it from existing studies that typically utilize only one of these technologies. Table 5 compares the success rates and distinct features of our proposed system with other relevant studies in the literature. Our proposed system's ability to combine BLE for local communication and LoRa for long-range transmission demonstrates its unprecedented capacity to ensure continuous communication during disasters. This hybrid approach bridges the gap between local and wide-area networks, achieving a 99% message delivery success rate.

To further distinguish and enhance the system, several potential future improvements and adaptations for different disaster types and geographic locations warrant consideration. For future development, enhanced sensor integration beyond the current GPS, accelerometer, temperature, vibration, and sound sensors could provide more granular and context-specific data. For instance, incorporating water level sensors for flood monitoring or air quality sensors for chemical spills could significantly improve situational awareness for targeted disaster responses. Furthermore, integration with UAVs (Unmanned Aerial Vehicles) as mobile LoRa-Router-Gateways or aerial relays could substantially extend the network's coverage, particularly in inaccessible areas, aligning with contemporary research in disaster communication. Implementing dynamic network configuration algorithms could enable the system to automatically optimize the LoRa mesh topology based on infrastructure damage and user distribution, enhancing overall network robustness. The integration of more advanced data analytics and artificial intelligence within the cloud infrastructure could facilitate predictive analysis, better prioritization of rescue efforts, and more intelligent resource allocation. Lastly, continued development of the control center UI to include features like interactive mapping with real-time tracking, multilingual support, and sophisticated alerting mechanisms would further improve usability for

TABLE 5. Comparison of disaster communication systems utilizing LoRa and LoRaWAN technologies.

Study	Technology Used	Success Rate	Unique Contribution
Vithayathil et al. (2021) [14]	LoRa	94%	Disaster rescue operations using LoRa and Wi-Fi
Macaraeg et al. (2020) [15]	LoRa Mesh Network	92%	Reliable off-grid communication with a modified AODV protocol
Sangrit et al. (2021) [16]	LoRa	91%	LoRa communication in mountainous regions
Baumgärtner et al. (2020) [20]	LoRa	89%	Location-aware communication for disaster scenarios
Jiang et al. (2021) [28]	Hybrid LoRa + ANT Mesh	>98% (Farm Deployment)	hybrid LPWAN mesh network
Shilpa et al. (2024) [30]	Hybrid LoRa + Wi-SUN	N/A	a hybrid LPWAN architecture
Ayele et al. (2018) [29]	BLE + LoRa (Potential)	N/A	adaptive MAC protocol for WSNs, potentially using BLE and LoRa, with a focus on range estimation
Humanity Lifeline	LoRa + LoRaWAN	99%	First hybrid system using both LoRa and LoRaWAN for resilient disaster communication

disaster management teams. The adaptability of the proposed system for different disaster scenarios is another crucial aspect for future exploration. For flood scenarios, prioritizing water level sensor integration and strategic deployment in vulnerable areas would be key. Expanding predefined emergency messages to include flood-specific needs would also be beneficial. In the context of earthquakes, while the current infrastructure-independent design is relevant, integrating seismic sensors (if feasible) could potentially provide early warning or infrastructure impact assessment. For wildfires, incorporating smoke and air quality sensors, alongside ruggedized hardware, would be essential, using LoRa's long-range capabilities for communication across vast affected areas. Adaptations for varying geographic locations would also be important. In densely populated urban areas, a higher density of LifeLinks and LoRa-Router-Gateways might be necessary, while in rural areas, maximizing the LoRa range and potentially utilizing UAV relays would be more critical. In mountainous regions, the peer-to-peer LoRa mesh network functionality becomes paramount, requiring strategic gateway deployment for multi-hop communication. For coastal areas, integrating sensors for water surge and wind speed, along with ensuring hardware waterproofing and resilience to harsh conditions, would be vital.

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